

Baseline and linear regression

A concise comparison on two synthetic datasets.

Dataset	Model	Training		Held-out test	
		MSE	MAE	MSE	MAE
Low noise	Mean	1.4221	1.0051	1.5476	1.1372
	Linear	0.0154	0.1012	0.0133	0.1011
Higher noise	Mean	1.6185	1.0422	1.8640	1.2140
	Linear	0.1706	0.3373	0.1474	0.3371

Table 1. Single-split results on synthetic data ($y = 2x + 1 + \epsilon$). Gaussian noise standard deviations are 0.15 (low) and 0.50 (higher). Each dataset uses 30 training and 10 test observations, with seed 42. MSE is mean squared error; MAE is mean absolute error. Lower values indicate smaller prediction errors.

Models. Mean predicts the training-target mean. Linear fits a slope and intercept using training data only.

Interpretation. The fitted line has lower held-out errors than the constant baseline on both datasets. These small synthetic examples illustrate a reporting workflow; they do not establish general performance.

Source. Values copied from `results.csv` at four decimal places. No uncertainty estimates are implied.